

Diabetes insipidus is a condition where the body loses too much fluid through urination, causing a significant risk of dangerous dehydration as well as a range of other illnesses and conditions. It is a rare disorder affecting the regulation of body fluid levels. People with diabetes insipidus produce excessive amounts of urine, resulting in frequent urination and thirst. However, the underlying cause of these two symptoms differs from types 1 and 2 diabetes.

9.4. Signs and Symptoms

- Polyuria or increased urination due to large volume of urine.
- Polydipsia or increased thirst due to excessive water loss and the need for its replacement in the tissues.
- Polyphagia or increased hunger due to a failure to utilize food for nourishment of the body.
- General weakness.
- Decreased resistance to infection.
- Decreased ability of wound healing due to a high blood sugar and poor fluid balance.
- Dehydration as a result of excessive water and electrolyte loss from the body, causing dry furred tongue and cracked lips.
- Ketosis or ketoacidosis i.e., accumulation of ketone bodies in the blood as a result of increased lipolysis. Breathing may be deep and rapid and the breath has acetone smell. Ketosis, if not controlled, may lead to coma and finally death.
- Rapid weight loss.
- Sleepiness
- Fatigue
- Degenerative changes in advanced cases include peripheral neuritis, retinitis, atherosclerosis with associated diseases of coronary arteries and vascular changes in kidneys causing nephropathy. Associated symptoms are failing or blurred vision, pain, numbness of the limbs and proteinuria.

9.5. Metabolic changes in Diabetes

Carbohydrate metabolism: Insulin lack produces and causes fundamental changes in carbohydrate metabolism which leads to hyperglycaemia. The primary is inability to regulate glucose in blood stream and the following changes in carbohydrate metabolism:

- Reduced entry and oxidation of glucose in muscle and tissues.
- Decreased formation of glycogen in liver.
- Decreased synthesis of fat as fats are mobilized as energy source instead of carbohydrates.
- Release of glucose into blood from increased breakdown of glycogen in liver.

Fat metabolism: Fat metabolism is also altered during diabetes mellitus. Due to the fact that the tissues are unable to oxidize sufficient quantities of glucose to meet energy needs, the body has to use fats as a source of energy. This results in mobilisation of large quantities of fat from adipose tissue and circulating as free fatty acids. This causes a considerable increase of fatty acid and triglycerides in the blood stream.

In the absence of normal carbohydrate metabolism, the liver oxidizes fatty acids on a larger scale producing large amounts of acetyl CoA which leads to the formation of large amounts of ketone bodies like acetoacetate, betahydroxy butyric acid and acetone. A metabolic acidosis develops known as ketosis which leads to coma in severe diabetes.

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